

# Notes on the Breidbart Basis

Intercept-and-Resend Attack on BB84 and  
Application to the IQT1 Quantum Token Protocol

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# 1 The BB84 Bases

The BB84 protocol uses two orthonormal bases of the qubit space  $\mathbb{C}^2$ :

$$\text{Computational basis (Z) : } |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1)$$

$$\text{Hadamard basis (X) : } |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (2)$$

The four BB84 states are  $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ . On the Bloch sphere, the  $Z$  basis lies along the  $z$ -axis and the  $X$  basis lies along the  $x$ -axis; they are separated by an angle of  $90^\circ$  on the sphere (equivalently,  $45^\circ$  in the Hilbert space picture).

## 2 The Breidbart Basis

### Definition 2.1 (Breidbart basis):

The *Breidbart basis*  $\mathcal{B} = \{|\phi_0\rangle, |\phi_1\rangle\}$  is defined by

$$|\phi_0\rangle = \cos\frac{\pi}{8} |0\rangle + \sin\frac{\pi}{8} |1\rangle \quad (3)$$

$$|\phi_1\rangle = -\sin\frac{\pi}{8} |0\rangle + \cos\frac{\pi}{8} |1\rangle \quad (4)$$

It is an orthonormal basis since  $\langle\phi_0|\phi_1\rangle = 0$  and  $\| |\phi_0\rangle \| = \| |\phi_1\rangle \| = 1$ .

### 2.1 Geometric interpretation

The Breidbart basis is obtained by rotating the  $Z$  basis by  $\pi/8 = 22.5^\circ$  in the Hilbert space, which corresponds to a rotation of  $\pi/4 = 45^\circ$  on the Bloch sphere. This places  $\mathcal{B}$  *exactly halfway* between the  $Z$  and  $X$  bases:

| Basis     | Bloch-sphere angle from $Z$ | Hilbert-space angle from $Z$ |
|-----------|-----------------------------|------------------------------|
| $Z$       | $0^\circ$                   | $0^\circ$                    |
| Breidbart | $45^\circ$                  | $22.5^\circ$                 |
| $X$       | $90^\circ$                  | $45^\circ$                   |

## 3 Intercept-and-Resend Attack in the Breidbart Basis

### 3.1 Setting up the attack

Bob (the bank, sender) prepares one of the four BB84 states and sends it as a single photon to Alice. Eve (= dishonest Alice in the IQT context) intercepts every photon and measures it in the Breidbart basis  $\mathcal{B}$ .

### 3.2 Computing the success probability

$Z$  basis states.

$$|\langle\phi_0|0\rangle|^2 = \cos^2\frac{\pi}{8} = \frac{2 + \sqrt{2}}{4} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \quad (5)$$

$$|\langle\phi_1|0\rangle|^2 = \sin^2\frac{\pi}{8} = \frac{2 - \sqrt{2}}{4} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \quad (6)$$

**X basis states.**

$$|\langle + | \phi_0 \rangle|^2 = \frac{1}{2}(1 + \sin \frac{\pi}{4}) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \quad (7)$$

The same value holds for all four BB84 states.

**Result 3.1 (Breidbart success and error probabilities):**

For *every* BB84 state:

$$p_{BE} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.8536 \quad e_{BE} = 1 - p_{BE} = \sin^2 \frac{\pi}{8} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 0.1464 \quad (8)$$

These are independent of which BB84 state was sent and which basis Bob used.

### 3.3 Why $\pi/8$ is optimal

The overall success probability for a general rotated basis at angle  $\theta/2$  is  $p_{\text{right}}(\theta) = \frac{1}{2} \cos^2(\theta/2) + \frac{1}{4}[1 + \sin \theta]$ . Setting  $dp_{\text{right}}/d\theta = 0$  gives  $\theta = \pi/4$ , i.e. Hilbert-space angle  $\pi/8$ . The Breidbart basis is the unique optimal intercept-and-resend basis for BB84.

## 4 QBER in the Two Scenarios

### 4.1 QKD scenario (Eve re-sends a photon)

Eve re-sends the Breidbart state; Bob measures in the correct BB84 basis. Agreement probability  $p_{AB} = p_{BE}^2 + e_{BE}^2 = 3/4$ , giving **QBER** = 0.25.

### 4.2 IQT scenario (Eve reports outcomes directly)

Eve records her Breidbart outcome and reports it as her token bit. Bob checks directly against  $t_k$ , giving

$$\text{QBER}_{\text{IQT}} = e_{BE} = \sin^2 \frac{\pi}{8} \approx 14.64\% \quad (9)$$

| Scenario                   | QBER                        | Value  |
|----------------------------|-----------------------------|--------|
| QKD (Eve re-sends photon)  | $1 - (p_{BE}^2 + e_{BE}^2)$ | 25%    |
| IQT (Eve reports directly) | $e_{BE} = \sin^2(\pi/8)$    | 14.64% |

## 5 Mutual Information

Eve's channel to each correct bit  $t_k$  is a BSC with crossover  $e_{BE}$ . Her mutual information per qubit is:

$$I_{BE} = 1 - h(e_{BE}) = 1 - h\left(\frac{1}{2} - \frac{1}{2\sqrt{2}}\right) \approx 0.399 \text{ bits/qubit} \quad (10)$$

The QKD key rate under full Breidbart attack ( $Q = 0.25$ ) is  $k = \max(1 - 2h(0.25), 0) = 0$ : no secret key can be extracted.

## 6 Application to the IQT1 Protocol

### 6.1 Protocol structure

In IQT1, Bob sends  $N$  BB84 single photons to Alice. After measurement and classical commitment, Alice presents at one of two spacelike-separated points  $Q_0$  or  $Q_1$ . The protocol algebra gives  $\tilde{d}_{b,k} = y_k$  always, so  $\Delta_b = \{k : y_k = u_k\}$  with  $|\Delta_b| \approx N/2$ .

### 6.2 QBER observed by the bank

$$\text{QBER}_b = e_{BE} = \sin^2 \frac{\pi}{8} \approx 14.64\% \quad \text{for } b \in \{0, 1\} \quad (11)$$

### 6.3 Maximum tolerable QBER $Q_{\text{tol}}$

Following the QKD Csiszár–Körner security condition, a positive secret key rate requires  $1 - 2h(q) > 0$ , i.e.  $h(q) < \frac{1}{2}$ .

#### Result 6.1 (Maximum tolerable QBER):

$Q_{\text{tol}}$  is the largest  $q$  such that  $2h(q) < 1$ , i.e. the solution of:

$$h(q) = \frac{1}{2} \implies \boxed{Q_{\text{tol}} \approx 11.00\%} \quad (12)$$

This is the security threshold from the BB84 structure shared by both QKD and IQT: any Eve with QBER above  $Q_{\text{tol}}$  cannot extract useful information.

The three key quantities satisfy:

$$\underbrace{Q_{\text{tol}}}_{\approx 11\%} < \underbrace{e_{BE}}_{\approx 14.64\%} < \underbrace{25\%}_{\text{individual attack}} \quad (13)$$

The margin  $e_{BE} - Q_{\text{tol}} \approx 3.64\%$  is a fundamental property of BB84 combined with the Breidbart geometry.

### 6.4 Bank's acceptance criterion and $\delta$

In the *ideal* IQT protocol, the bank sets  $\gamma_{\text{err}} = 0$  (zero tolerance). In practice, one sets  $\gamma_{\text{err}} \in (0, Q_{\text{tol}})$ . We write:

$$\gamma_{\text{err}} = e_{BE} - \delta \quad \text{where } \delta \in (0, e_{BE}) \quad (14)$$

$\delta$  is the gap between Eve's true error rate  $e_{BE}$  and the bank's threshold. The larger  $\delta$ , the stricter the bank.

### 6.5 Forging probability and minimum token size

The bank accepts Eve's token if her empirical error count falls below  $\gamma_{\text{err}} \cdot n = (e_{BE} - \delta) \cdot n$ . Since Eve's errors follow  $\text{Bin}(n, e_{BE})$  with mean  $e_{BE} \cdot n$ , the forging probability is a *lower tail* probability:

$$\boxed{P_{\text{forge}}(\delta, n) = P(\text{Bin}(n, e_{BE}) \leq (e_{BE} - \delta) \cdot n)} \quad (15)$$

For large  $n$ , the central limit theorem gives the normal approximation:

$$P_{\text{forge}}(\delta, n) \approx \Phi(z), \quad z = \frac{-\delta\sqrt{n}}{\sqrt{e_{BE}(1 - e_{BE})}} \quad (16)$$

where  $\Phi$  is the standard normal CDF and  $z < 0$  since we are in the lower tail ( $\gamma_{\text{err}} < e_{BE}$ ). Substituting  $z$  and squaring:

**Result 6.2 (Minimum token size  $n_{\min}$ ):**

$$n_{\min}(\delta) = \frac{[\Phi^{-1}(P_{\text{forge}}^{\text{target}})]^2 \cdot e_{BE}(1 - e_{BE})}{\delta^2} \quad (17)$$

$n_{\min}$  scales as  $1/\delta^2$ : halving  $\delta$  quadruples the required token size. At  $\delta = e_{BE} - Q_{\text{tol}} = 3.64\%$  and  $P_{\text{forge}} = 10^{-9}$  (so  $|\Phi^{-1}| \approx 6.0$ ):

$$n_{\min} = \frac{(6.0)^2 \times 0.1250}{(0.0364)^2} \approx 3,400 \quad (18)$$

*Ideal IQT* ( $\gamma_{\text{err}} = 0$ ,  $\delta = e_{BE}$ ):  $n_{\min} \approx 210$  — the large gap means very few photons suffice.

**Remark 6.1 ( $Q_{\text{tol}}$  and  $\gamma_{\text{err}}$  are the same in ideal IQT):**

In the ideal protocol, the bank tolerates zero errors:  $\gamma_{\text{err}} = 0$ . As one allows noise,  $\gamma_{\text{err}}$  can be raised up to  $Q_{\text{tol}} \approx 11\%$  — beyond which the QKD security condition  $2h(q) < 1$  is violated. So  $Q_{\text{tol}}$  is both the QKD security threshold *and* the maximum permissible  $\gamma_{\text{err}}$ .

## 7 Unforgeability Bound: Equation (D1)

**Result 7.1 (Lemma 1 / eq. (D1), Kent et al. [1]):**

$$\varepsilon_{\text{unf}} = \left( \frac{1}{2} + \frac{1}{2\sqrt{2}} \right)^N = p_{BE}^N \quad (\text{D1})$$

The base is exactly the per-qubit Breidbart success probability: the Breidbart attack is optimal, and its per-qubit success rate directly gives this bound.

Selected values:

| $N$  | $\varepsilon_{\text{unf}} = p_{BE}^N$ |
|------|---------------------------------------|
| 100  | $\approx 10^{-7}$                     |
| 500  | $\approx 10^{-34}$                    |
| 1000 | $\approx 10^{-69}$                    |
| 2000 | $\approx 10^{-138}$                   |

## 8 Summary

| Quantity                   | Formula                                                                       | Value                                | Nature                        |
|----------------------------|-------------------------------------------------------------------------------|--------------------------------------|-------------------------------|
| $p_{BE}$                   | $\frac{1}{2} + \frac{1}{2\sqrt{2}}$                                           | 0.8536                               | Born's rule                   |
| $e_{BE}$                   | $\sin^2(\pi/8)$                                                               | 14.64%                               | Born's rule (IQT QBER)        |
| $Q_{\text{tol}}$           | $h(q) = \frac{1}{2}$                                                          | 11.00%                               | QKD security condition        |
| margin                     | $e_{BE} - Q_{\text{tol}}$                                                     | 3.64%                                | fundamental gap               |
| $\delta$                   | $e_{BE} - \gamma_{\text{err}}$                                                | $(0, e_{BE})$                        | bank's strictness             |
| $\gamma_{\text{err}}$      | $e_{BE} - \delta$                                                             | $[0, Q_{\text{tol}}]$                | chosen (protocol)             |
| $n_{\text{min}}$           | $[\Phi^{-1}(P_{\text{forge}}^{\text{target}})]^2 e_{BE}(1 - e_{BE})/\delta^2$ | $\approx 3,400$ at $\delta = 3.64\%$ | derived                       |
| $\varepsilon_{\text{unf}}$ | $p_{BE}^N$                                                                    | $\sim 10^{-69}$ at $N = 1000$        | eq. (D1) — full quantum bound |

The chain of inequalities governing security:

$$\underbrace{0}_{\text{ideal } \gamma_{\text{err}}} \leq \underbrace{\gamma_{\text{err}}}_{\text{chosen}} \leq \underbrace{Q_{\text{tol}}}_{11\%} < \underbrace{e_{BE}}_{14.64\%} \implies P_{\text{forge}} \text{ negligible for } n \geq n_{\text{min}}(\delta) \quad (19)$$

## References

- [1] A. Kent, D. Lowndes, D. Pitalúa-García, J. Rarity, *Practical quantum tokens without quantum memories and experimental tests*, arXiv:2104.11717v4 (2022).